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Anomalous pulse-angle and phase dependence of Hahn's electron spin echo and multiple-quantum echoes in a photoinduced spin-correlated radical pair

Jau Tang[a] and Marion C. Thurnauer
Chemistry Division, Argonne National Laboratory, Argonne, Illinois 60439

Atsushi Kubo
Department of Chemistry, Kyoto University, Sakyoku Kyoto 606, Japan

Hideyuki Hara and Asako Kawamori
Faculty of Science, Kwansei Gakuin University, Nishinomiya 662, Japan

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Formation of two- and three-pulse electron spin echoes and their anomalous pulse-angle and phase dependence in a photoinduced spin-correlated radical pair are examined. Analytical formulae for the unusual Hahn's spin-echo signals and the two-pulse double-quantum echo are derived in terms of the pulse angle, microwave phase, and the time intervals between the laser pulse and the microwave pulses. Such anomalies are shown to be present in both cases of singlet and triplet precursors. Phase dependence and formation of the three-pulse stimulated echo and multiple-quantum echoes are also examined. © *1997 American Institute of Physics.* [S0021-9606(97)00318-8]

I. INTRODUCTION

Unusual spin polarization for a photoinduced spin-correlated radical pair has been widely known (for reviews, see Refs. 1–3). Pulsed electron paramagnetic resonance (EPR) techniques have often been used in the study of isolated radical and triplet systems (Ref. 4, and references therein). Electron spin-echo envelope modulation in these systems is usually caused by anisotropic hyperfine interactions or nuclear quadruple interactions, and the depth of the modulation is often shallow. In our recent theoretical work of spin dynamics for a photoinduced spin-correlated radical pair system with strong pulses,[5–7] some interesting features were predicted, such as the out-of-phase echo and deep envelope modulation due to electron–electron dipolar or exchange interaction. We have shown that the modulation frequencies are free of hyperfine inhomogeneity and, therefore, these frequencies provide a simple way to determine the distance information between the unpaired electron spins. The direct measurement of the dipolar coupling, however, is not feasible by transient cw EPR or even quantum-beat experiments[8,9] that usually produce unresolved polarized spectra broadened by hyperfine interactions. The abnormal phase effect has also been noted independently by Salikhov *et al.*[10] Experimental observation of such an out-of-phase echo has been first reported by Moenne-Loccoz, Heathcote, and Evans,[11] and more recently by Dzuba, Gast, and Hoff,[12] who also detected deep modulation induced by electron–electron dipolar interactions in bacterial photosynthetic systems. This technique has been recently applied to photosystem I.[13] Another kind of interesting out-of-phase echo effect with weak pulses was first reported much earlier by Thurnauer and Norris,[14] which is caused by sequential electron transfer during microwave pulses. Such an effect requires a sudden change of local fields (or the spin-quantization axis) due to electron transfer processes that occur between pulses. Therefore, their mechanism is different from the phase anomaly discussed here for the strong pulse regime.

In our earlier study,[5] only a singlet precursor was assumed and the simplest Hahn's 90°-τ-180° pulse sequence was considered with an identical phase for both two-microwave pulses. In this work, we consider both cases of a singlet and a triplet precursor. In addition, we also report other unusual properties for a photoinduced radical pair, such as the anomalous pulse-angle dependence of the electron spin-echo (ESE) signals and the effect of the phase of the microwave pulses. More general formulae with a closed-form expression will be presented for both singlet and triplet precursor cases. Phase memory decay will also be considered for the zero-quantum coherence between two nearly degenerate spin levels and the ordinary single-quantum coherence for the transverse magnetization. It will be shown that these anomalies occur for both cases with a singlet and a triplet precursor. They are caused by the photoinduced non-Boltzmann population and zero-quantum coherence, and the singlet–triplet mixing mechanism. Each electron spin for the radical pair precesses at a different rate as a result of different g factors or hyperfine interactions with adjacent nuclear spins.

In addition to these special features for a two-pulse sequence, we will also examine the formation of an additional double-quantum echo that does not occur in the ordinary situation in nuclear magnetic resonance (NMR) or EPR spectroscopy. The intensity has unusual pulse-angle dependence and vanishes for a 90°–180° pulse sequence. We will also investigate the stimulated echo and other unusual echoes in a three-pulse sequence. Study of the anomalous effects of these multiple-quantum echoes in the photoinduced radical

[a]Author to whom correspondence should be addressed. Electronic mail: TANG@ANLCHM.CHM.ANL.GOV

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pairs in natural and artificial photosynthesis can provide useful structural information that is not readily available by ordinary transient cw EPR measurements.

II. THEORY

We will first establish in Sec. II A the spin Hamiltonian, its eigenstates, and eigenenergies for a spin-correlated radical pair. The matrix representations for the propagator and the pulse operator of arbitrary pulse angle and phase will be given in Sec. II B. Time evolution of the density matrix for a two-pulse sequence of flash$-t-P1-\tau-$P2$-\tau$ will be evaluated in Sec. II C. The double-quantum echo and its dependence on a phase shift will be illustrated in Sec. II D. In Sec. II E the stimulated echo and multiple-quantum echoes using three microwave pulses will also be examined.

A. The spin Hamiltonian

In the treatment of the ESE of a radical pair system in a magnetic field with a strength B_0, it is more convenient to use the rotating frame representation at microwave frequency Ω. The spin Hamiltonian is given by[5]

$$H=(g_1\beta B_0-\Omega)S_{1z}+(g_2\beta B_0-\Omega)S_{2z}+S_{1z}\sum_\alpha A_{1\alpha}I_{\alpha z}+S_{2z}\sum_\alpha A_{2\alpha}I_{\alpha z}+J(\tfrac{1}{2}-2\mathbf{S}_1\cdot\mathbf{S}_2)+\frac{D}{2}(3\cos^2\theta-1)\times[S_z^2-\tfrac{1}{3}S(S+1)], \quad (1)$$

where g_1 and g_2 are g factors of the radical spin $\mathbf{S}_1$ and $\mathbf{S}_2$, $A_{1\alpha}$ and $A_{2\alpha}$ are their hyperfine coupling constants with nuclear spin $\mathbf{I}_\alpha$, J is the exchange interaction, and D is the electron–electron dipolar interaction strength. For convenience $\hbar\equiv1$ is used. If the microwave field is present, one should include the rf Hamiltonian $\omega_1(S_{1x}+S_{2x})$. The dipolar interaction will be assumed to have axial symmetry that makes an angle θ with respect to the external magnetic field. An ensemble average for the angular distributions for the dipolar orientation (θ) should be taken for a powder sample.

The singlet–triplet mixing is caused by differences in g values and hyperfine interactions for $\mathbf{S}_1$ and $\mathbf{S}_2$. The off-diagonal elements $\langle S|H|T_0\rangle$ and $\langle T_0|H|S\rangle$ are generally nonzero. Because $\mathbf{S}_1$ and $\mathbf{S}_2$ experience a different local field, they precess at a different frequency. Therefore, the initial population in the singlet (for a singlet precursor) or the triplet (for a triplet precursor) begins to oscillate in time between the singlet and triplet states. Because of the singlet–triplet mixing, $|S\rangle$ and $|T_0\rangle$ are not eigenstates of H. Instead, two middle eigenstates $|\Phi_a\rangle$ and $|\Phi_b\rangle$ are a mixture of $|S\rangle$ and $|T_0\rangle$ as given by

$$|\Phi_a\rangle=\cos\varphi|S\rangle+\sin\varphi|T_0\rangle,$$

$$|\Phi_b\rangle=-\sin\varphi|S\rangle+\cos\varphi|T_0\rangle;$$

$$\varphi\equiv\tfrac{1}{2}\tan^{-1}\left[\frac{(g_1-g_2)\beta B_0+\Sigma_\alpha(A_{1\alpha}-A_{2\alpha})m_\alpha}{2J+(D/3)(3\cos^2\theta-1)}\right]. \quad (2)$$

Here, m_α is the nuclear magnetic quantum number. A Gaussian distribution of the nuclear hyperfine fields with a different second moment is assumed for each of $\Sigma_\alpha A_{1\alpha}m_\alpha$ and $\Sigma_\alpha A_{2\alpha}m_\alpha$. The eigenvalues of the spin Hamiltonian are given by

$$E_1=\langle T_+|H|T_+\rangle=\frac{1}{2}(g_1+g_2)\beta B_0-\Omega+\frac{1}{2}\sum_\alpha(A_{1\alpha}+A_{2\alpha})m_\alpha+\frac{D}{6}(3\cos^2\theta-1),$$

$$E_2=\langle\Phi_a|H|\Phi_a\rangle=J-\frac{D}{6}(3\cos^2\theta-1)+\sqrt{[J+(D/6)(3\cos^2\theta-1)]^2+\tfrac{1}{4}\left[(g_1-g_2)\beta B_0+\sum_\alpha(A_{1\alpha}-A_{2\alpha})m_\alpha\right]^2},$$

$$E_3=\langle\Phi_b|H|\Phi_b\rangle=J-\frac{D}{6}(3\cos^2\theta-1)-\sqrt{[J+(D/6)(3\cos^2\theta-1)]^2+\tfrac{1}{4}\left[(g_1-g_2)\beta B_0+\sum_\alpha(A_{1\alpha}-A_{2\alpha})m_\alpha\right]^2},$$

$$E_4=\langle T_-|H|T_-\rangle=-\frac{1}{2}(g_1+g_2)\beta B_0+\Omega-\frac{1}{2}\sum_\alpha(A_{1\alpha}+A_{2\alpha})m_\alpha+\frac{D}{6}(3\cos^2\theta-1). \quad (3)$$

B. Evolution of the density matrix after the laser pulse

In this treatment, we will use the matrix representation with the basis set of $|T_+\rangle$, $|S\rangle$, $|T_0\rangle$, and $|T_-\rangle$. The propagator $\exp(-iHt)$ can be expressed by

$$e^{-iHt}=\begin{pmatrix} e^{-iE_1t} & 0 & 0 & 0\\ 0 & \cos^2\varphi e^{-iE_2t}+\sin^2\varphi e^{-iE_3t} & \tfrac{1}{2}\sin 2\varphi(e^{-iE_2t}-e^{-iE_3t}) & 0\\ 0 & \tfrac{1}{2}\sin 2\varphi(e^{-iE_2t}-e^{-iE_3t}) & \sin^2\varphi e^{-iE_2t}+\cos^2\varphi e^{-iE_3t} & 0\\ 0 & 0 & 0 & e^{-iE_4t}\end{pmatrix}. \quad (4)$$

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The density matrix $\rho(t)$ at a time t after the laser pulse is given by $\exp(-iHt)\rho(0)\exp(iHt)$. Using the basis set of $|T_+\rangle$, $|S\rangle$, $|T_0\rangle$, and $|T_-\rangle$ and assuming a singlet precursor, one can show $\rho(t)$ as[5]

$$\rho(t)=\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1-G & F & 0 \\ 0 & F^* & G & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \tag{5a}$$

or

$$\begin{aligned} &\rho_{ss}(t)=1-G(t), \\ &\rho_{T_0T_0}(t)=G(t), \\ &\rho_{ST_0}(t)=F(t), \\ &G(t)\equiv \tfrac{1}{2}\sin^2 2\varphi(1-e^{-t/T_{23}}\cos E_{23}t), \\ &F(t)\equiv \tfrac{1}{2}\sin 2\varphi[\cos 2\varphi(1-e^{-t/T_{23}}\cos E_{23}t) \\ &\qquad +ie^{-t/T_{23}}\sin E_{23}t], \end{aligned} \tag{5b}$$

where $E_{23}=E_2-E_3$. Here, we have included the effects of phase-memory decay with a time constant T_{23} for the photo-induced zero-quantum coherence between $|\Phi_a\rangle$ and $|\Phi_b\rangle$. For a triplet precursor, $\rho(t)$ is then given by

$$\rho(t)=\frac{1}{3}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & G & -F & 0 \\ 0 & -F^* & 1-G & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{6}$$

Using the matrix representation for S_x and S_y

$$S_x=\frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \tag{7}$$

$$S_y=\frac{i}{\sqrt{2}}\begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix},$$

we have obtained in the following the rotation operator $P_{\zeta,\xi}$ for an ideal microwave pulse with the pulse angle ζ and the phase ξ, as defined by $P_{\zeta,\xi}=\exp[-i\zeta(S_x\cos\xi-S_y\sin\xi)]$,

$$P_{\zeta,\xi}=\begin{pmatrix} \frac{1}{2}(1+\cos\zeta) & 0 & -\frac{i}{\sqrt{2}}e^{i\xi}\sin\zeta & -\frac{1}{2}e^{2i\xi}(1-\cos\zeta) \\ 0 & 1 & 0 & 0 \\ -\frac{i}{\sqrt{2}}e^{-i\xi}\sin\zeta & 0 & \cos\zeta & -\frac{i}{\sqrt{2}}e^{i\xi}\sin\zeta \\ -\frac{1}{2}e^{-2i\xi}(1-\cos\zeta) & 0 & -\frac{i}{\sqrt{2}}e^{-i\xi}\sin\zeta & \frac{1}{2}(1+\cos\zeta) \end{pmatrix}. \tag{8}$$

C. Electron spin echo following a flash–t–$P1$–τ–$P2$–τ pulse sequence

Using the matrix representations derived earlier for the microwave pulse operator $P_{\zeta,\xi}$ in Eq. (8), the propagator $\exp(-iHt)$ in Eq. (4), the spin operators S_x and S_y in Eq. (7), and the density matrices in Eqs. (5)–(6) at a time t after the flash for either a singlet or triplet precursor, one can now evaluate the amplitude for the electron spin echo following a flash–t–$P_{\zeta,\xi}$–τ–P_{180}–τ pulse sequence. Here, the pulse angle ζ and the phase ξ of the first pulse $P_{\zeta,\xi}$ can be arbitrary, whereas the pulse angle for the second pulse P_{180} is 180°. The ESE signals $S_x(t,\tau)$ and $S_y(t,\tau)$, corresponding to the transverse magnetization at a time τ after the second pulse, are given by

$$\begin{aligned} S_x(t,\tau)&=\mathrm{Tr}(S_xe^{-iH\tau}P_{180,0}e^{-iH\tau}P_{\zeta,\xi}\rho(t) \\ &\quad\times P^+_{\zeta,\xi}e^{iH\tau}P^+_{180,0}e^{iH\tau}), \\ S_y(t,\tau)&=\mathrm{Tr}(S_ye^{-iH\tau}P_{180,0}e^{-iH\tau}P_{\zeta,\xi}\rho(t) \\ &\quad\times P^+_{\zeta,\xi}e^{iH\tau}P^+_{180,0}e^{iH\tau}). \end{aligned} \tag{9}$$

We have obtained the echo amplitude

$$\begin{aligned} &S_x(t,\tau)=\cos\xi\,[A_1(t,\tau)+A_2(t,\tau)], \\ &S_y(t,\tau)=\sin\xi\,[A_1(t,\tau)+A_2(t,\tau)], \\ &A_1(t,\tau)\equiv\sin\zeta\sin^2 2\varphi\cos 2\varphi e^{-2\tau/T_2}(\cos E_{23}\tau-1) \\ &\qquad\times[e^{-t/T_{23}}\sin E_{23}t\cos(E_{12}-E_{34})\tau+\cos 2\varphi(1 \\ &\qquad-e^{-t/T_{23}}\cos E_{23}t)\sin(E_{12}-E_{34})\tau], \end{aligned}$$

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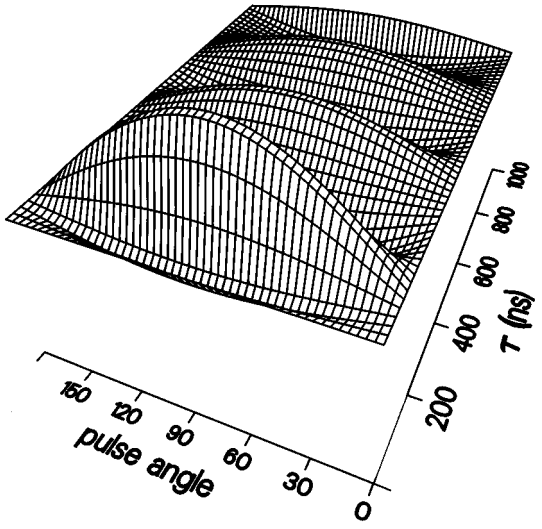


FIG. 1. Two-dimensional signals of $A_1(t,\tau)$ in Eq. (10) with $t=1$ μs showing $\sin\zeta$ dependence on the pulse angle. The other dimension along the τ axis shows the deep modulation due to electron–electron dipolar interactions. Here, the flash–t–$P_{45,0}$–τ–$P_{180,0}$–τ sequence is used.

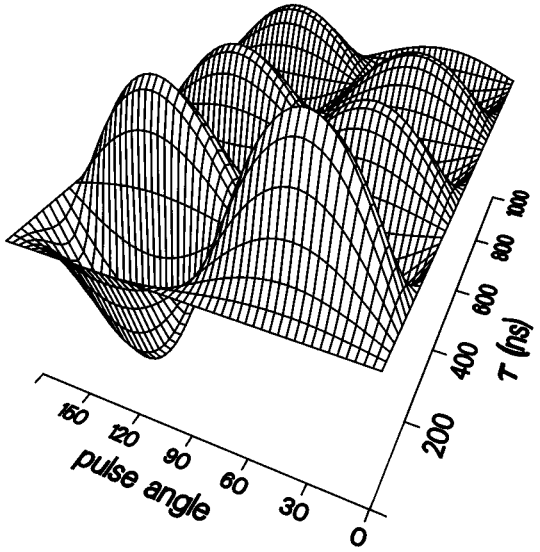


FIG. 2. Two-dimensional signals of $A_2(t,\tau)$ in Eq. (10) with $t=1$ μs showing $\sin 2\zeta$ dependence on the pulse angle. It arises from $\langle T_0|\rho(t)|T_0\rangle$, whereas $A_1(t,\tau)$ with $\sin\zeta$ dependence arises from $\langle S|\rho(t)|T_0\rangle$. In this display, a reduction of the amplitude for $A_2(t,\tau)$ by 10 is used for a better comparison with $A_1(t,\tau)$ in Fig. 1.

$$A_2(t,\tau)\equiv\tfrac{1}{2}\sin 2\zeta\,\sin^2 2\varphi e^{-2\tau/T_2}(e^{-t/T_{23}}\cos E_{23}t-1)\times[\cos 2\varphi\,\sin E_{23}\tau\cos(E_{12}-E_{34})\tau+(\sin^2 2\varphi+\cos^2 2\varphi\,\cos E_{23}\tau)\sin(E_{12}-E_{34})\tau]. \quad (10)$$

Here, we include the transverse relaxation with a time constant T_2 for four single-quantum transitions and T_{23} for the phase-memory decay of zero-quantum coherence.

Using two microwave pulses along the x axis, it is found that spin magnetization is refocused along the x axis, contrary to the ordinary in-phase Hahn echo for Boltzmann population, where an echo appears along the y axis. Such an anomaly is independent of the pulse angle ζ. The formula obtained in our earlier treatment[5] is a special case of Eq. (10) with $\zeta=90°$, $\xi=0$, and no phase-memory decay. If the phase ξ for the first pulse P_1 is 0, i.e., along the x axis, one can easily see from the above equation that $S_y(t,\tau)=0$. This phenomenon with an out-of-phase echo has been confirmed in recent experiments by Evans' group[11] and Hoff's group.[12] From Eq. (3) one can show that the modulation frequency is given by $E_{12}-E_{34}=2D(3\cos^2\theta-1)/3-2J$. It depends only on the dipolar and exchange couplings and is independent of hyperfine inhomogeneity.

In addition to the out-of-phase anomaly for Hahn's spin echo, there is unusual dependence of the echo height on the pulse angle ζ. From Eq. (10) one can see the echo consists of two terms A_1 and A_2. A_1 depends on $\sin\zeta$ with a maximum for a 90° pulse whereas A_2 depends on $\sin 2\zeta$ with extremes for a 45° or 135° pulse. This phenomenon is also different from the ordinary Hahn echo that has a maximum with a 90° pulse. This pulse-angle anomaly is illustrated in Figs. 1–3. In all the simulations of this work, $J=0$ G, $D=1.7$ G and $(g_1-g_2)\beta B_0=6.8$ G are used. The square root of the second moment of the hyperfine field at each electron spin is set at 10 and 7 G, respectively. These values are typical for photosystem I.[13] To obtain a smoother curve with sufficient ensemble averaging, 50 different angles for the dipolar fields and 100 discrete values with a Gaussian distribution for each hyperfine field are used. A two-dimensional display of the out-of-phase echo is shown in Figs. 1 and 2. Both illustrations exhibit along the τ axis a strong deep ESE envelope modulation due to electron–electron dipolar interactions. Such modulation provides a simple and direct measurement of the dipolar coupling and the distance between the unpaired electron spins. Along the other axis, two sources of pulse-angle dependence are illustrated, one showing $\sin\zeta$ dependence and the other showing $\sin 2\zeta$ dependence. The overall out-of-phase echo amplitude with individual contri-

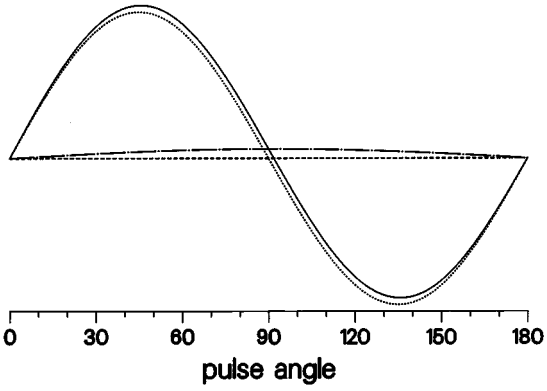


FIG. 3. The overall dependence of the echo height (solid line) on the pulse-angle ζ containing two different sources with a $\sin\zeta$-dependent term (chain–dot line) and a $\sin 2\zeta$-dependent term (dot line). The zero-base line is represented by the dash line. Here, $t=1$ μs and $\tau=140$ ns.

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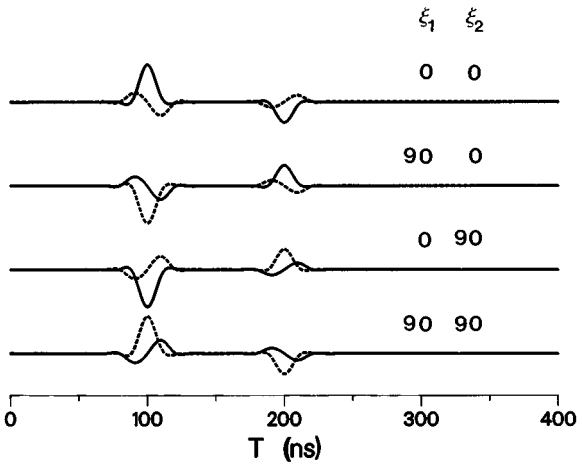


FIG. 4. The T dependence of $S_x(t,\tau,T)$ (solid line) and $S_y(t,\tau,T)$ (dash line) showing two echoes. The flash$-t-P_{\zeta,\xi 1}-\tau-P_{2\zeta,\xi 2}-T$ sequence with $t=\tau=100$ ns is used. The phase dependence on two pulses (ξ_1 and ξ_2) indicates that the second echo at 2τ is due to manifestation of double-quantum coherence that evolves between $P1$ and $P2$, whereas the first echo is due to single-quantum coherence in the evolution period τ.

bution for the pulse-angle dependence is illustrated in Fig. 3. The $\sin\zeta$-dependent term comes from $\langle S|\rho(t)|T_0\rangle$ but the $\sin 2\zeta$-dependent term comes from $\langle S|\rho(t)|S\rangle$ and $\langle T_0|\rho(t)|T_0\rangle$. One can show that $\langle S|\rho(t)|T_0\rangle$ transforms as a vector by the rotation of a microwave pulse and has a $\sin\zeta$ dependence, whereas $\langle T_0|\rho(t)|T_0\rangle$ transforms as a second-order tensor and has a $\sin 2\zeta$ dependence. Because of the ensemble averaging of multiple nuclear spin configurations, the oscillation of the populations between these two states can be highly damped. The ensemble average of $F(t)$ [the off-diagonal term $\langle S|\rho(t)|T_0\rangle$] is usually smaller than that of $G(t)$ [the diagonal term $\langle T_0|\rho(t)|T_0\rangle$], and this results in a weaker contribution from the $\sin\zeta$-dependent term. The strongest out-of-phase echo, therefore, would occur for the pulse angle ζ near 45° or 135° as illustrated in Fig. 4.

Similarly, for a triplet precursor we have obtained the spin-echo signals as

$$S_x(t,\tau)=-\tfrac{1}{3}\cos\xi\,[A_1(t,\tau)+A_2(t,\tau)],$$
$$S_y(t,\tau)=-\tfrac{1}{3}\sin\xi\,[A_1(t,\tau)+A_2(t,\tau)], \tag{11}$$

where $A_1(t,\tau)$ and $A_2(t,\tau)$ are defined earlier in Eq. (10). The above results show that for the triplet–precursor case one has the similar out-of-phase effect. The echo amplitude has an opposite polarity and the intensity is reduced by a factor of 3. The polarity of the echo provides a simple tool to distinguish between a singlet or triplet precursor.

D. The double-quantum echo with two microwave pulses in a flash$-t-P1-\tau-P2-2\tau$ pulse sequence

We will examine the double-quantum echo with two microwave pulses in a flash$-t-P_{\zeta,\xi}-\tau-P_{2\zeta,0}-T$ pulse sequence where $t=\tau$ and $T=2\tau$. After a straightforward but lengthy calculation, we have obtained a closed-form expression for the amplitude of the second echo that appears at $T=2\tau$

$$\begin{aligned}S_x(t,\tau)=&-\tfrac{1}{8}\cos 2\xi\,\sin 2\zeta(1-\cos 2\zeta)^2\\&\times\sin^2 2\varphi(1-\cos E_{23}\tau)[(1+\tfrac{1}{2}\sin^2 2\varphi)\\&\times\cos E_{23}\tau\,\sin(E_{12}-E_{34})\tau\\&+\cos 2\varphi\,\sin E_{23}\tau\,\cos(E_{12}-E_{34})\tau].\end{aligned} \tag{12}$$

The above equation with $\cos 2\xi$ dependence on the microwave phase ξ indicates that this echo is a double-quantum echo. The pulse-angle dependence of the double-quantum echo is given by Eq. (12) as $\sin 2\zeta(1-\cos 2\zeta)^2$, which has a maximum for ζ at about 1.211 rad, or 69.29°. This echo disappears, however, for a 90°–180° pulse sequence.

In our earlier work,[7] we have pointed out that double-quantum coherence between $|T_+\rangle$ and $|T_-\rangle$ can be excited by the first microwave pulse $P1$ because of the initial photo-induced zero quantum coherence between $|\Phi_a\rangle$ and $|\Phi_b\rangle$, in contrast to the ordinary multiple-quantum NMR techniques[15–18] that require at least two rf pulses for excitation. Excitation of multiple-quantum coherence in an optical four-level system by a single pulse was also considered by Warren and Zewail,[19] however, the details are different from the system considered here. In a recent pulsed EPR experiment by Dzuba *et al.*,[20] the presence of the double-quantum coherence was confirmed. They observed in the detection period T an echo at a time $T=\tau$ and a much weaker echo at $T=2\tau$ that is attributed to double-quantum coherence. The second echo at $T=2\tau$ arises from the special condition of $t=\tau$. The photoinduced zero-quantum coherence is transformed by the first pulse into single- and double-quantum coherence that evolve during $P1$ and $P2$. Then, the second pulse $P2$ converts the coherence into detectable single-quantum signals. Both echoes are produced by refocused transverse magnetization of single-quantum signals that are directly detectable by an EPR cavity. As illustrated in Fig. 4, the formation of the second echo at 2τ and its characteristic phase dependence are the manifestation of double-quantum coherence that evolves between $P1$ and $P2$. The amplitude of the second echo is inverted by a phase shift in ξ_1 of the first microwave pulse by 90°. The reason is that $P1$ converts the zero-quantum coherence into double-quantum coherence that experiences a phase shift by $2\Delta\xi_1$ as the phase of $P1$ is shifted by $\Delta\xi_1$. Although $P1$ can convert S_z^2 polarization into double-quantum coherence in the period between $P1$ and $P2$, this mechanism would not contribute to the formation of the unusual double-quantum echo at $T=2\tau$ after $P2$. The reason is that as the time t between the flash and $P1$ increases and differs from τ, our simulations indicate that the unusual double-quantum echo diminishes. A change in t would not change the S_z^2 polarization significantly, which represents diagonal elements with a slow decay. However, a small change in t can easily mismatch the echo conditions for the phase of the initial zero-quantum coherence generated by the flash. Therefore, the observation of such a two-pulse double-quantum echo at $T=2\tau$ with $t=\tau$ is strong evidence for the presence of the photoinduced zero-quantum coherence. The response of the echo shape to the phase ξ_2 of the second pulse is rather different. For the second echo at 2τ,

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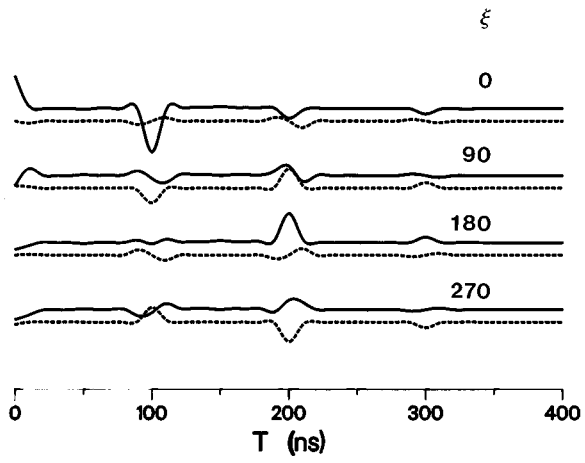


FIG. 5. The envelope modulation of the three-pulse stimulated echo $S_x(t,\tau,T)$ and $S_y(t,\tau,T)$ using the flash$-t-P_{\zeta,\xi}-\tau-P_{2\zeta,\xi}-T-P_{90}-\tau$ pulse sequence, where $t=\tau=100$ ns and $\zeta=67.5°$. Four visible echoes are present in the evolution period T at $T=0$, 100, 200, and 300 ns. Their phase dependence can be used to distinguish the origins of these echoes.

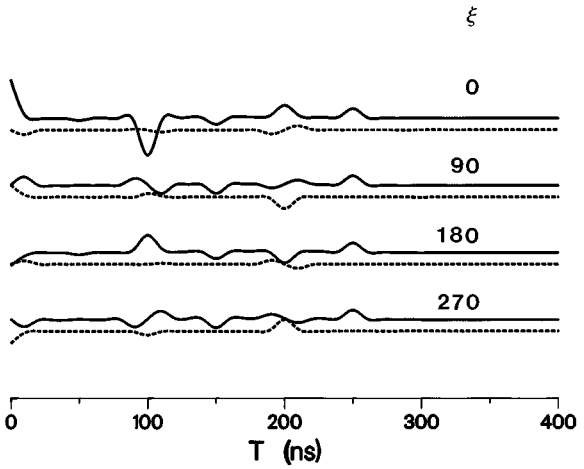


FIG. 6. Formation of echoes in the T dependence of $S_x(t,\tau,T)$ and $S_y(t,\tau,T)$, where $t=150$ ns, $\tau=100$ ns, and $\zeta=67.5°$. The echoes at $T=150$ and 250 ns are zero-quantum echoes, whereas the echo at $T=200$ ns is a single-quantum echo.

$P2$ transforms the double-quantum coherence into detectable single-quantum coherence. Therefore, the net phase change of the echo is only $\Delta\xi_2$ if one shifts the phase of $P2$ by $\Delta\xi_2$. The first echo, however, is inverted by a 90° phase shift on the second pulse. This indicates its origin from the single-quantum coherence that exists before $P2$. $\langle\Phi_a|\rho|T_-\rangle$ and $\langle\Phi_b|\rho|T_-\rangle$ are interexchanged into $\langle\Phi_a|\rho|T_+\rangle$ and $\langle\Phi_b|\rho|T_+\rangle$ by $P2$. This switching results in a net phase change for the first echo by $2\Delta\xi_2$ if $P2$ is shifted by $\Delta\xi_2$.

E. The stimulated echo and multiple-quantum echoes with three microwave pulses

We will now examine the stimulated echo and multiple-quantum echo with three microwave pulses in a flash$-t-P1-\tau-P2-T-P3-\tau$ pulse sequence. The intensity is given by

$$\begin{aligned} S_x(t,\tau,T) &= \mathrm{Tr}(S_x e^{-iH\tau}P_3 e^{-iHT}P_2 e^{-iH\tau}P_1\rho(t) \\ &\quad\times P_1^+ e^{iH\tau}P_2^+ e^{iHT}P_3^+ e^{iH\tau}), \\ S_y(t,\tau,T) &= \mathrm{Tr}(S_y e^{-iH\tau}P_3 e^{-iHT}P_2 e^{-iH\tau}P_1\rho(t) \\ &\quad\times P_1^+ e^{iH\tau}P_2^+ e^{iHT}P_3^+ e^{iH\tau}). \end{aligned} \tag{13}$$

Using the matrix representations for various relevant operators obtained in Eqs. (4)–(8), one can evaluate the echo amplitude for such a multiple-pulse sequence and derive an analytical formula. However, because the expression is too lengthy to be recorded here, only numerical simulations will be presented in this work. Unlike the two-pulse echo that exhibits deep modulation by electron–electron dipolar or exchange interactions, we have found from simulation that the T dependence of $S_x(t,\tau,T)$ and $S_y(t,\tau,T)$ is rather smooth and the envelope of the stimulated echo does not exhibit such oscillatory behavior. Instead, several echoes appear. Unlike the familiar case in NMR or EPR with initial Boltzmann population, the appearance and the phase dependence of these echoes in a photoinduced radical pair are unusual. For illustration as shown in Fig. 5, let us first consider the flash$-t-P_{\zeta,\xi}-\tau-P_{2\zeta,\xi}-T-P_{90}-\tau$ pulse sequence, where $t=\tau=100$ ns and $\zeta=67.5°$ (or $3\pi/8$). Four visible echoes in the evolution period T are present at $T=0$, 100, 200, and 300 ns. For a pure m-quantum echo ($m=0$, 1, and 2 for the biradical system), its phase dependence is $m\Delta\xi$ as one shifts the phase of the first two pulses by $\Delta\xi$. The echo at 300 ns is a single-quantum echo because of its simple $\Delta\xi$ dependence. The other echoes, however, are not purely zero-, single-, or double-quantum echoes because of their more complicated phase dependence. With a different setting of t and τ, where $t=150$ ns and $\tau=100$ ns, many more echoes are present as illustrated by Fig. 6. They appear at $T=0$, 50, 100, 150, 200, and 250 ns. The echoes at $T=150$ and 250 are not affected by a phase shift in ξ, therefore, they are zero-quantum echoes. The echo at $T=200$ ns has simple $\Delta\xi$ dependence and is a single-quantum echo. These echoes of multiple-quantum coherence are formed in the evolution period (so called in multiple-quantum NMR literature). This evolution period allows nonobservable multiple-quantum coherence to evolve before conversion into observable single-quantum coherence in the detection period. These echoes are different from the double-quantum echo that appears in the detection period discussed in Sec. II D. The formation of these multiple-quantum echoes and their intensities depend also in a different way on microwave resonant conditions. A field scan with boxcar detection set at these zero, single, and double echoes can provide useful information to complement other measurements using echo-envelope modulations. Therefore, the phase-cycled two- and three-pulse techniques proposed in this work to observe selectively a specific zero-, single-, or double-quantum echo in combination with a field scan and envelope modulation can complement other methods to pro-

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vide more direct measurements of dipolar and exchange interactions as well as isotropic/anisotropic hyperfine and Zeeman interactions.

III. DISCUSSION AND CONCLUSIONS

In this theoretical study of spin dynamics of a photoinduced spin-correlated radical pair system, we have examined the anomalous effects of microwave pulse angle and phase on the two-pulse echoes, the three-pulse stimulated echo, and multiple-quantum echoes. We have derived in Eq. (10) two-dimensional analytical expressions for a general two-pulse spin-echo sequence where the pulse angle and phase for the first pulse are arbitrary. In addition, we have also considered the cases of both a singlet and triplet precursor. Phase-memory decays for the zero-quantum (T_{23}) and single-quantum coherence (T_2) is also included in our closed-form formulae. The echo amplitude for the triplet precursor is found in Eq. (11) to have an opposite sign and a reduced height (1/3) as compared to that for the singlet precursor case. Therefore, the polarity of the echo can be used to distinguish whether the photoinduced radical pair originates from a singlet or a triplet precursor. The origin of the pulse-angle anomaly and its two different sources are analyzed. Experimental examination of the ζ dependence can help distinguish these two different contributions. Ernst *et al.*[21] noted a maximal signal with a 45° pulse for the NMR FID (free induction decay) signals of a system with an initial population at nonequilibrium. Similar observations in EPR FID signals from polarized, interacting radical pairs were reported by Levstein and van Willigen[22] and Hasharoni *et al.*[23] These studies, however, do not consider the photoinduced zero-quantum coherence that exists initially between $|\Phi_a\rangle$ and $|\Phi_b\rangle$ in the photoinduced spin-correlated radical pair. As illustrated in Fig. 3, because the sin 2ζ term is more dominant than the sin ζ term, the maximal EPR signals would occur at around 45°. Anomalous pulse-angle dependence can occur in a different situation of partial microwave excitation of $S>1/2$ spin systems as demonstrated by Isoya *et al.*[24] We have also illustrated the out-of-phase effect with the flash$-t-P_{\zeta,0}-\tau-P_{180,0}-T$. Such an out-of-phase anomaly is shown to be independent of the pulse angle ζ. The modulation frequency for the out-of-phase echo depends only on the dipolar and exchange couplings and is independent of hyperfine inhomogeneity. The measurement of the modulation frequency for the out-of-phase echo provides a simple and useful method to determine directly the distance between the unpaired electron spins.

In addition to the anomalous Hahn echo at $T=\tau$ in a two-pulse sequence where t is set at τ, an additional echo appears at $T=2\tau$ and is shown in Eq. (12) to be a double-quantum echo with 2ξ dependence in phase. Such an echo does not occur in the ordinary situation in NMR or EPR spectroscopy. The presence of this double-quantum echo is a result of spin dynamics involved in a photoinduced radical pair. These results agree with the experimental observation by Dzuba *et al.*[20] Its intensity has unusual pulse-angle dependence as given by $\sin 2\zeta(1-\cos 2\zeta)^2$ that has a maximum for ζ at about 69°. This echo disappears for a 90°–180° pulse sequence.

We have also examined the three-pulse stimulated echo with a flash$-t-P1-\tau-P2-T-P3-\tau$ pulse sequence. Unlike the two-pulse echo that exhibits deep modulation by dipolar interactions, the envelope of the stimulated echo does not exhibit such oscillatory behavior in T. Instead, many echoes appear in the evolution period T of the flash$-t-P_{\zeta,\xi}-\tau-P_{2\zeta,\xi}-T-P_{90}-\tau$ pulse sequence. These echoes have different phase dependence and are the result of zero-, single-, and double-quantum coherence. A specific multiple-quantum echo can be selected out by the phase-cycling technique. A field scan or envelope modulation of each individual multiple-quantum echo can provide useful information to complement other techniques in the measurement of dipolar, exchange, hyperfine, and Zeeman interactions of the photoinduced radical pairs in natural and artificial photosynthesis. As also illustrated recently by the work of Saxena and Freed,[25] they were able to measure distances by double-quantum signals.

Subsequent charge separation during the pulse sequence can also produce an out-of-phase echo for weak pulses.[14] We have shown that a slow electron transfer process can produce significant effects on transient polarized cw EPR spectra.[26] In this work, strong microwave pulses are assumed. For a weak microwave pulse, the transformation matrix $P_{\zeta,\xi}$ cannot be expressed in a closed analytical form. A numerical approach with matrix diagonalization is required for evaluation of the echo signals. Theoretical treatment of these factors and the effects of sequential electron transfer processes for pulsed EPR will be included in future work.

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